Reuse Water System – Runbook

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Document control

Item	Detail
Document	Reuse Water System – Runbook
Version	3.6
Applies to firmware	`FSM_WashingMachineSystem.ino` and the `GreyWater*` files, firmware 3.6 (3.6: four new screen fonts chosen by the household, the earlier ones removed; 3.5: view-only user `viewer` that sees only Overview and Water use; 3.4.1: shorter screen messages, full screen brightness; 3.4: minimum flow to save, screen font chosen in Settings; 3.3.1: alerts tried again for 30 min during an outage; 3.3: screen messages in short pages that explain what is happening and what to do, minutes left, more reliable phone alerts; 3.2: new name Reuse Water System, start-up animation, Helvetica Bold screen font, larger tank icon, buzzer on D1; 3.1: Spanish and English, chosen in Settings; 3.0: safety task, float-switch rules, code split into modules)
Intended audience	Installers, operators and maintenance personnel
Related documents	`README.md` (firmware change summary); Reuse Water System – Quick Card

Purpose and scope

This runbook defines how to install, operate, configure, troubleshoot and maintain the Reuse Water System controller. It covers the controller firmware and the hardware connected directly to it. The washing machine, the plumbing design and the mains electrical installation are outside its scope.

Safety precautions

Isolation: switch off and lock out the valve and pump supply before working on any wiring. The Stop button is a software control and does not isolate power.

Electrical work: if the valves or pump run on mains voltage, the SSR load wiring must be installed by a qualified electrician to local regulations, with overcurrent protection and an RCD/GFCI.

Environment: mount the controller in an enclosure rated for damp locations, and keep low-voltage wiring separated from mains wiring.

Hygiene: grey water contains detergent residue and bacteria. Wear gloves when servicing the tank, sensors and valves. Stored water is not potable.

Supervision: do not leave the system unattended during commissioning or after a firmware change.

Quick start

For everyday use you only need the two buttons and the screen. After the controller is switched on, the screen shows a dog chasing a butterfly for about 4 seconds, then `REUSE WATER` / System ready, then `READY`.

1. **Check the screen** shows `READY` and *Start washer › Press START*.
2. **Load the washing machine** and start its program as usual.
3. **Press START** (the button labelled START). The screen changes to `WASH 1 OF 4`.
4. **Let it run.** The top line shows the current step, `WASH 1 OF 4` to `SAVE 4 OF 4`. The message below comes in short parts that change every 3 seconds, for example *Reusing water › 25 min left*. In the last step it shows when the system finishes.
5. **Done.** The screen shows `DONE` with *Reused 26 L › Saved 30 L*, then returns to `READY`.

To cancel at any time, press **STOP** (the button labelled STOP). The screen shows `STOPPED` for 4 seconds.

On a phone connected to the home Wi-Fi, open `http://reusewater.local` to see the same status, start or stop the cycle, and check how much water you have saved (section 11).

If the screen shows	It means	What to do
A dog chasing a butterfly	The controller has just been switched on (about 4 s)	Nothing; START skips it
READY / Start washer › Press START	Waiting for you	Start the washing machine, then press START
WASH 1 OF 4 … SAVE 4 OF 4	Cycle in progress	Nothing; wait
Reusing water	The machine is filling with water saved in the tank	Nothing
25 min left	About how long the cycle still needs, learnt from earlier washes (from the second wash)	Nothing
Filling tank › Ends in 18:42	Rinse water is being stored in the tank for the next wash	Nothing

If the screen shows	It means	What to do
DONE / Reused 26 L › Saved 30 L	Cycle finished; tank water used, and rinse water stored for next time	Nothing
STOPPED / Cycle cancelled	You pressed STOP	Press START to begin again
No water flow › Check washer › Call installer	A step has waited too long for water	Check the washing machine is running; if it is, press STOP and call your installer (section 8)
Blank screen	No power, or a display fault	Check power, then see section 8

Button colors are a recommendation; label your buttons to match.

1. Overview

The controller runs a 4-step wash cycle (Fill, Drain, Rinse, Drain), reuses grey water from a storage tank for the first fill, and saves the rinse water back to the tank. It runs on an Arduino Nano ESP32, drives five solid-state relays (SSRs), reads two flow sensors and two tank level switches, and reports status on a 128x32 OLED.

Components

Group	Device	What it does	Board pin	Part / type
Controller	Arduino Nano ESP32	Runs the program	—	ABX00083 (ESP32-S3, 3.3 V logic)
Operator	START button	Starts a cycle	D11	Momentary push button, normally open
Operator	STOP button	Cancels a cycle	D10	Momentary push button, normally open
Operator	OLED screen	Shows status and progress	A4 (SDA), A5 (SCL)	SSD1306, 128x32, I2C 0x3C
Operator	Buzzer (optional)	Beeps when a cycle ends and on alarms	D1 / TX0	Active buzzer module (with its own driver)
Sensors	Tap flow sensor	Detects water entering the machine	D2	YF-B1
Sensors	Drain flow sensor	Detects water leaving the machine	D3	YF-B10
Sensors	Tank low-level switch	Water at the low mark	D9	Float switch

Group	Device	What it does	Board pin	Part / type
Sensors	Tank high-level switch	Water at the high mark (tank full)	D8	Float switch
Actuators	Tap valve	Tap water supply	D6	Solenoid valve via SSR
Actuators	Drain valve	Drain path from the machine	D4	Solenoid valve via SSR
Actuators	Fill valve	Sends drain water into the tank	D5	Solenoid valve via SSR
Actuators	Pump	Moves tank water to the machine	D7	Pump via SSR
Actuators	Tank-wash valve	Washes the tank during the tank-water fill	D0 / RX0	Solenoid valve via SSR

Wash cycle summary

1. **Fill (1/4):** the machine fills with tap water, or with tank water when the tank has water and the machine is demanding water. The tank-wash valve pulses every 10 s while tank water is used.
2. **Drain (2/4):** soapy water is drained, not saved.
3. **Rinse (3/4):** the machine fills with tap water.
4. **Drain (4/4):** rinse water is saved to the tank until it is full, but only while the drain flow is above the *Minimum flow to save* (default 0.5 L/min, from 3.4). When the flow drops to that value or below, the fill valve closes and the drain opens at once, so the last trickle goes to the drain; a new pump-out (a spin) is saved again once it has flowed above the limit for 2 s. This step lasts 20 minutes, then the system returns to Idle.

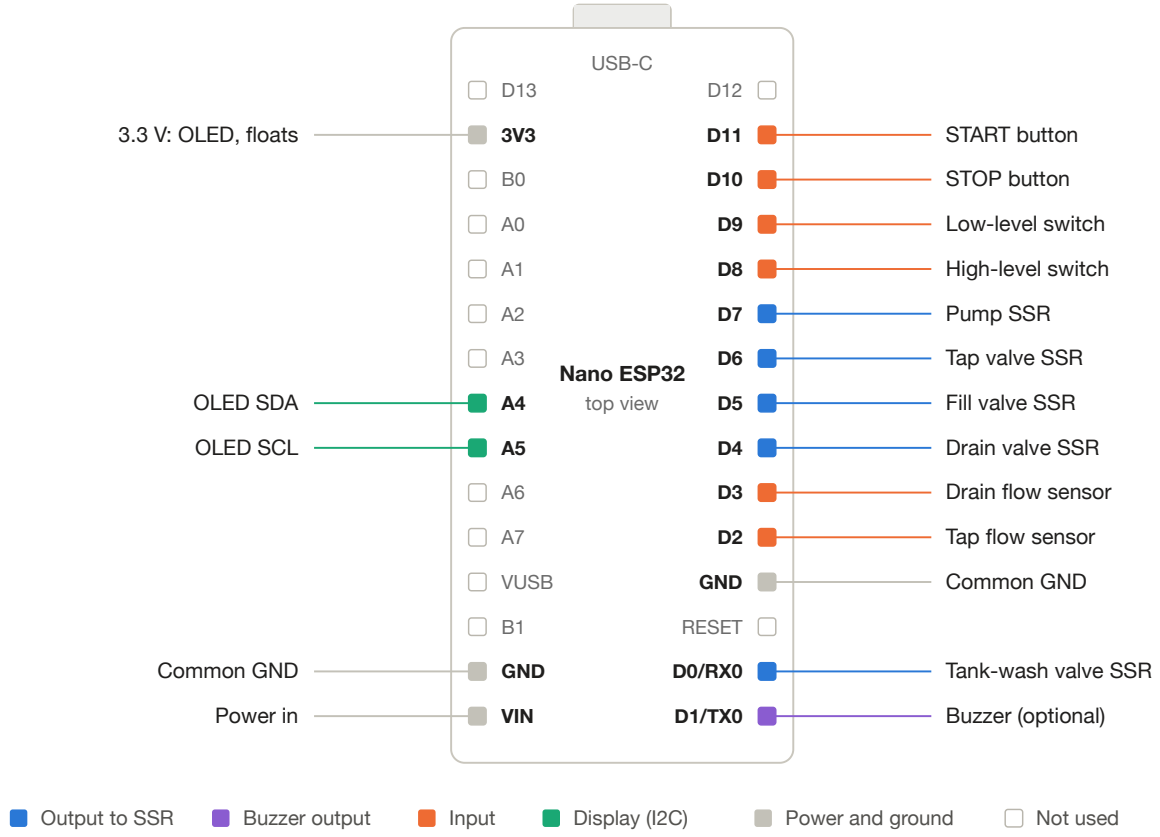
Source files (from firmware 3.0): `FSM_WashingMachineSystem.ino` (setup and loop), `GreyWaterSystem` (wash cycle), `GreyWaterOutputs` and `GreyWaterTask` (outputs, float switches, safety task), `GreyWaterSensors`, `GreyWaterDisplay`, `GreyWaterDiagnostics`, `GreyWaterSettings`, `GreyWaterNet`, `Ntfy.h` (phone alert message, from 3.3), `FlowMeter.h`, `TankLevel.h`, the screen files `BootAnimation.h`, the screen fonts `FontHyperlegible13.h` … `FontSchoolbook10.h` (from 3.6, see `licenses/README.md`), and the app page `web/dashboard.html` with its Spanish texts `web/es.json`. Libraries: Adafruit GFX, Adafruit SSD1306, AgileStateMachine, WiFiManager.

2. Pinout configuration

All pins are defined in `GreyWaterSystem.h` and set up in `setup()`. Outputs are active HIGH (`ssrON = HIGH`); buttons are active LOW with internal pull-ups.

Board pinout

Sensors, buttons, valves and buzzer use the D0–D11 header; the display uses A4 and A5



Arduino Nano ESP32 pinout · pins used by this system

Filled squares are the pins in use. Match each pin by the label printed on the board, not by its position, because the drawing is not to scale.

Board pin	ESP32-S3 GPIO	Connect to	Type	Reads / drives	Firmware name
Inputs					
D2	GPIO5	Tap flow sensor signal	Pulse input (interrupt)	Pulses while water flows	`TAP_PIN`

Board pin	ESP32-S3 GPIO	Connect to	Type	Reads / drives	Firmware name
D3	GPIO6	Drain flow sensor signal	Pulse input (interrupt)	Pulses while water flows	`DRAIN_PIN`
D8	GPIO17	Tank high-level switch	Digital input	HIGH = water at the high mark	`IN_HIGHLEVEL`
D9	GPIO18	Tank low-level switch	Digital input	HIGH = water at the low mark	`IN_LOWLEVEL`
D10	GPIO21	STOP button (other side to GND)	Input, internal pull-up	LOW = pressed	`STOP_BUTTON`
D11	GPIO38	START button (other side to GND)	Input, internal pull-up	LOW = pressed	`START_BUTTON`
Outputs					
D0 / RX0	GPIO44	Tank-wash valve SSR, input +	Digital output (shared with UART0 RX)	HIGH = valve on	`OUT_WASH_EV`
D1 / TX0	GPIO43	Buzzer module signal (optional)	Digital output (shared with UART0 TX)	HIGH = beep	`BUZZER_PIN`
D4	GPIO7	Drain valve SSR, input +	Digital output	HIGH = valve on	`OUT_DRAIN_EV`
D5	GPIO8	Fill valve SSR, input +	Digital output	HIGH = valve on	`OUT_FILL_EV`
D6	GPIO9	Tap valve SSR, input +	Digital output	HIGH = valve on	`OUT_TAP_EV`
D7	GPIO10	Pump SSR, input +	Digital output	HIGH = pump on	`OUT_MOTOR`
Display					
A4	GPIO11	OLED SDA	I2C data, 400 kHz	—	`Wire`
A5	GPIO12	OLED SCL	I2C clock, 400 kHz	—	`Wire`
Power					
3V3	—	OLED VCC, level-switch common and buzzer module VCC	3.3 V supply out	—	—

Board pin	ESP32-S3 GPIO	Connect to	Type	Reads / drives	Firmware name
GND	—	Common ground: buttons, sensors, OLED, buzzer, every SSR input –, and the 5 V supply –	Ground	—	—
VIN	—	Controller supply (see the board datasheet for the allowed voltage)	Power in	—	—

Wiring requirements

SSR inputs: connect each output pin to the SSR's DC input (+) and GND to (–). The board must not drive solenoids or the motor directly.

Level switches: D8 and D9 use plain `INPUT`, so each switch needs an external pull-down (or pull-up, with the logic inverted in `getWaterLevelTank()`). A floating input gives random tank levels.

Buttons: wire between the pin and GND; no external resistor is needed.

Flow sensors: power them at the voltage the sensor and the board's inputs both accept. The Nano ESP32 inputs are 3.3 V only and not 5 V tolerant. If a sensor is powered from 5 V, use the diode and 10 kΩ pull-up shown in the wiring diagrams on D2 and D3. Do not use a resistor divider: with the sensor's own pull-up it can leave the signal just above the pin's switching level, so pulses drop out.

Pin D0 (tank-wash valve): D0 is also the board's RX0 serial pin. The sketch's `Serial` uses the USB port, so D0 is free, but do not use `Serial0` in the same sketch. Fit a 10 kΩ resistor from D0 to GND so the SSR stays firmly off while the board starts. Always write `D0` in the code, not a bare `0`.

Pin D1 (buzzer, optional, from firmware 3.2): D1 is also the board's TX0 serial pin, so do not use `Serial0`. Use an active buzzer module (one that beeps when its input is HIGH): signal to D1, VCC to 3V3, GND to GND. The chip sends its own start-up messages on TX0 before the firmware runs, so the buzzer may chirp briefly at power-on or after a reset; this is normal. The first line of `setup()` drives D1 LOW. Always write `D1` in the code, not a bare `1`. Until 3.1 the buzzer was on D12.

Boot state: `allSSROff()` writes LOW to all five outputs in `setup()`, so every valve and the motor start off.

Pin numbers follow the Arduino labels printed on the board (Arduino IDE: **Tools > Pin Numbering > By Arduino pin (default)**). The GPIO column is for reference only. Source: Arduino Nano ESP32

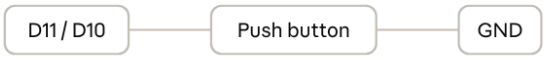
full pinout.

Wiring diagrams

Each panel shows one type of device; wire every device of that type the same way, using the pin from the pinout table.

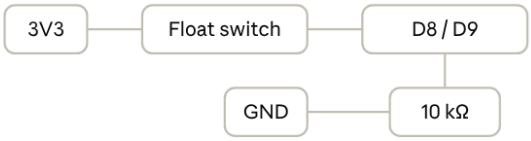
Sensors and buttons run at 3.3 V; only the SSR load side carries valve power

START and STOP buttons (D11, D10)



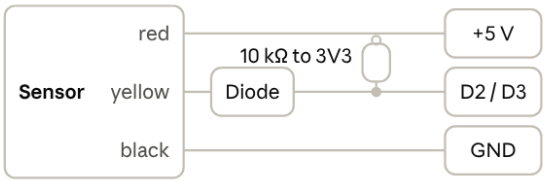
Uses the internal pull-up: no resistor needed.
Pressed = LOW. Use normally open buttons.

Tank level switches (D8, D9)



Float up: switch closed, pin reads HIGH.
The 10 kΩ pull-down holds the pin LOW when open.

Flow sensors (D2, D3)



Diode + 10 kΩ pull-up: a clean 0–3.3 V signal.
Diode cathode to the sensor; colors are typical.

Valves and pump via SSR (D0, D4–D7)



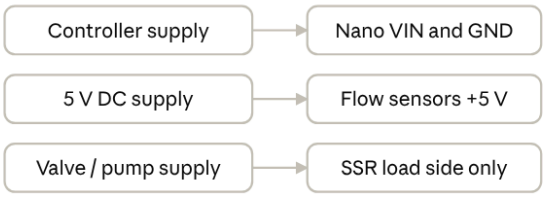
AC valves: AC-output SSR on the load supply.
DC valves: DC-output SSR and a flyback diode.

OLED display (A4, A5)



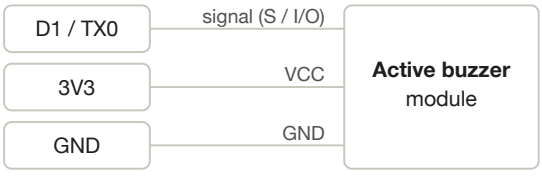
Address 0x3C; most modules include pull-ups.

Power supplies and ground



Join low-voltage GNDs; keep load supply separate.

Buzzer, optional (D1)



HIGH = beep. A short chirp at power-on is normal (TX0).

Choosing the buzzer

Active buzzer module with its own driver (it beeps by itself when its input is HIGH).

A passive buzzer or a bare piezo will not work: the firmware only switches the pin.

Switch it on in the app: Settings → Buzzer.

Wiring diagrams · one circuit per device type

Device	Wire 1	Wire 2	Wire 3	Extra parts
START / STOP button	Pin D11 / D10	GND	—	None
Level switch (×2)	3V3	Pin D8 / D9	—	10 kΩ from the pin to GND
Flow sensor (×2)	Red: +5 V supply	Black: GND	Yellow: pin D2 / D3 through a Schottky diode (cathode to the sensor)	10 kΩ from the pin to 3V3; optional 1–10 nF from the pin to GND
SSR (×5)	Input +: pin D0 or D4–D7	Input –: GND	—	Load side in series with the valve or pump and its supply
OLED	VCC: 3V3	GND: GND	SDA: A4, SCL: A5	None
Buzzer (optional)	Signal: pin D1 / TX0	VCC: 3V3	GND: GND	None (active buzzer module)

Choose SSRs whose control input switches reliably at 3.3 V (for example, a 3–32 V DC input). Use an AC-output SSR for AC valves and pumps, and a DC-output SSR with a flyback diode across the coil for DC valves.

3. Outputs and relay states

From firmware 3.0, only `GreyWaterOutputs` switches the outputs. The active step requests one of the modes below every 250 ms (`OPERATION_DELAY`), and a separate safety task checks the floats and the pump every 20 ms, whatever the rest of the program is doing (for example while a phone loads the app).

What the system is doing	When	Tap valve (D6)	Drain valve (D4)	Fill valve (D5)	Pump (D7)	Tank-wash valve (D0)	Firmware function
Starting up	Power-on	Off	Off	Off	Off	Off	`allSSROff()`
Waiting / not saving	Idle, Run, step 4 with tank full	On	On	Off	Off	Off	`noSavingWater()`

What the system is doing	When	Tap valve (D6)	Drain valve (D4)	Fill valve (D5)	Pump (D7)	Tank-wash valve (D0)	Firmware function
Washing the tank manually	Idle, after Wash tank on the dashboard	On	On	Off	Off	On until the tank is full	`idleStateLogic()`
Filling with tap water	Step 1 (tap), step 3	On	On	Off	Off	Off	`fillWashingMachineWithTapWater()`
Filling with tank water	Step 1 (tank), while the machine draws water	Off	On	Off	On; off within 0.3 s of the machine closing its inlet	Pulsed: 2 s every 10 s	`fillWashingMachineWithTankWater()`
Draining soapy water	Step 2	On	On	Off	Off	Off	`drainSoapyWater()`
Saving water to the tank	Step 4, tank not full, drain flow above the minimum flow to save	On	Off	On	Off	Off	`checkAndFillWaterTank()`

Pump protection (from firmware 2.5): the pump runs only while water is actually flowing.

- **Fast cut-off:** if no tap-sensor pulse arrives for 0.3 s, the pump stops and the tap valve opens at once. That happens when the machine closes its inlet, or when the pump delivers nothing (air lock, blocked intake, empty pipe). The pump gets 1 s after starting for water to flow.
- **No restart into a closed inlet:** after a cut-off, the pump waits for the machine's next draw.
- **Alert:** after 3 starts with no water in one cycle, the controller sends a *tank pump not pumping* alert.
- **Hardware limit:** firmware cannot remove the last 0.3 s. A pump with a built-in pressure switch, or a bypass back to the tank, protects it fully.

Interlocks (from firmware 3.0): applied every 20 ms in every state.

- The fill valve (D5) and the tank-wash valve (D0) never open while the high float is wet. In step 4 the rinse water then goes to the drain, and saving stays off for the rest of that step.
- The pump (D7) never runs while the low float is dry; the tap supplies the machine instead.
- The pump cut-off above keeps working even if the main program is busy.

Tank level decoding

From firmware 3.0, the safety task reads both switches every 20 ms. Each switch reports its safe reading after 0.2 s (high float wet, low float dry) and the other reading only after 2 s, so a float bobbing at the mark does not switch valves on and off:

Low switch D9	High switch D8	Level	OLED tank icon
LOW	LOW	low	Empty (outline with half-way marks)
HIGH	LOW	medium	Half full
HIGH	HIGH	high	Full
LOW	HIGH	high: the tank is treated as full and the pump is not used. After 3 s it is reported as a switch fault (alert *check level switches*, app *Check float switches*); see section 8	Full

Tank-wash timing

While `fillWashingMachineWithTankWater()` runs, the wash valve opens immediately, stays open for `TANK_WASH_OPEN_TIME` (2 s), and reopens every `TANK_WASH_PERIOD` (10 s). The timer restarts on every state entry. Switching to tap water closes the valve; switching back to tank water in the same state continues the same 10 s cycle.

4. Buttons and operator controls

The system has two momentary buttons: Start begins a cycle from Idle, and Stop ends a cycle from any other state. Both are read every `loop()` pass by `updateInputs()` with a 20 ms debounce (`DEBOUNCE_DELAY`).

Button	Pin	Press it when	What happens	Screen shows	Ignored when
START	D11	The screen shows `READY` and the washing machine program is running	The cycle begins; step 1 starts after 250 ms. During the start-up animation it also ends the animation	`STARTING`, then `1/4 WASH`	A cycle is already running
STOP	D10	You want to cancel the cycle	The system returns to Idle and resets all timers	`STOPPED` for 4 s, then `READY`	The system is already in Idle

Operational notes

- **Start in any other state:** ignored. A running cycle cannot be restarted without pressing Stop first.
- **Stop in Idle:** ignored.
- **Stop paths:** from Run and Fill 1/4 the state machine's own transition goes to Idle. From Drain 2/4, Rinse 3/4 and Drain 4/4, `loop()` also forces a reset with `resetFSMToIdle()`. Both end in the same Idle entry.
- **Buttons act while held:** the debounce reports the button as pressed for as long as it is held, not once per press. Holding Start when a cycle ends starts the next cycle straight away.
- **Both buttons at once:** do not press both buttons simultaneously. The resulting state depends on the order in which transitions are evaluated.
- **Stop is not an emergency stop.** In Idle the tap and drain SSRs stay on (`noSavingWater()`). To make the installation safe, isolate the SSR and valve supply (see Safety precautions).

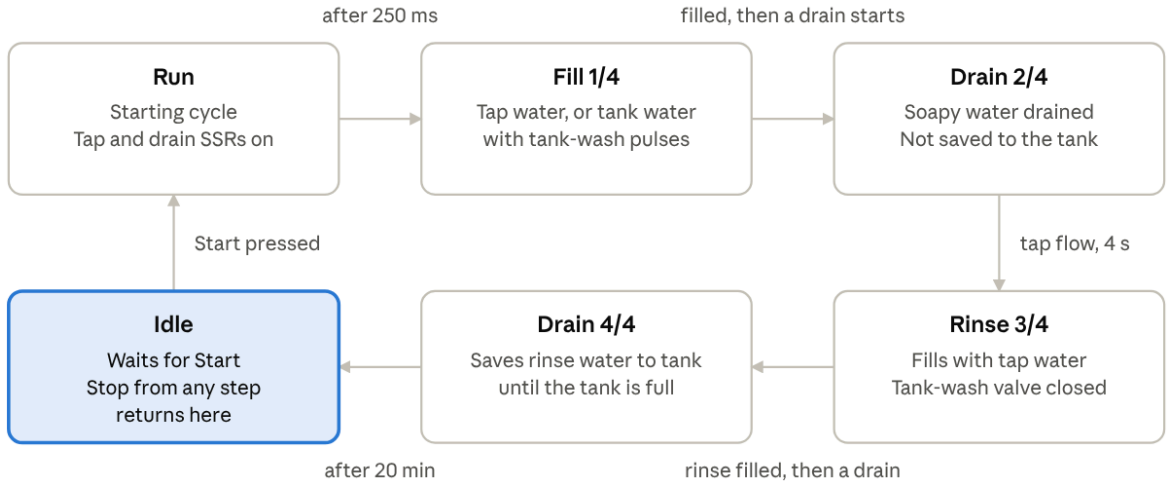
Standard operating procedure

1. Check that the OLED shows `READY` / Press START.
2. Load the washing machine and start its program.
3. Press Start. The header changes to `1/4 WASH`.
4. No operator action is required while the cycle runs. Progress shows in the header (`1/4` to `4/4`); in step 4 a countdown shows the time left.
5. When the OLED shows `DONE` / Cycle complete, the system is back in Idle.
6. To abort, press Stop at any time.

5. Operating procedure

Pressing Start runs the four steps in order; each of steps 1–3 ends only when the flow sensors confirm what the washing machine is doing, and step 4 ends after 20 minutes.

Start runs four steps, then the system returns to Idle



State machine · 6 states, Stop returns to Idle from every step

The diagram reads clockwise from Run. "Flow, 4 s" means the flow rate stayed above the Flow threshold (default 1 L/min) for the flow detection time, `EDGE_DETECTION_DELAY` (4 s).

State	On entry	Every 250 ms	Next state	Exit condition
Idle	Clears buttons, cycle flags, edge flags, flow counters, tank-wash timer; shows the result notice	`noSavingWater()`	Run	Start pressed
Run	Clears any notice	`noSavingWater()`	Fill 1/4	250 ms elapsed
Fill 1/4	Resets flow counters and tank-wash timer	`checkAndFillWashingMachine()`: tap water if the tank is low or no demand yet, else tank water with wash pulses	Drain 2/4	The machine is filled (took at least the Minimum fill, default 3 L, and its inlet closed), then a drain starts (`filled1`). A drain before or during the fill does not count.

State	On entry	Every 250 ms	Next state	Exit condition
Drain 2/4	Resets flow counters	`drainSoapyWater()`	Rinse 3/4	Tap flow seen for 4 s: the machine starts its rinse fill (`drained1`)
Rinse 3/4	Resets flow counters	`fillWashingMachineWithTapWater()`	Drain 4/4	The rinse fill is complete (Minimum fill taken, inlet closed), then a new drain starts: the rinse drain (`filled2`)
Drain 4/4	Resets flow counters; starts the countdown	`checkAndFillWaterTank()`: save to tank while low or medium, else not saving	Idle	`TRANSITION_DELAY` (20 min) elapsed

Steps 1–3 have no timeout. If the expected flow never arrives, the step waits until Stop is pressed (section 8).

The controller follows the washing machine; it does not command it. Start the machine's own program, then press Start, and keep the machine's program length in line with `TRANSITION_DELAY`.

6. OLED display guide

The screen shows the step in the header, two live indicators on the right, and a one-line message below. From firmware 3.3 a message can have up to five short pages that change every 3 seconds (`DISPLAY_PAGE_MS`): what is happening, what it means, and what to do or how long it takes. In the tables below the pages are separated by ›. One line holds only about 15 characters in the large font, so pages say more without making the text smaller.



Element	Reads	Meaning
Header text	`WASH 1 OF 4` etc.	Current step (from 3.3 in words: `WASH 1 OF 4`, `DRAIN 2 OF 4`, `RINSE 3 OF 4`, `SAVE 4 OF 4`; until 3.2 `1/4 WASH`). Built-in 6 px font, at most 16 characters

Element	Reads	Meaning
Tank icon	17 × 10 px box: empty with half-way marks / half / full	Tank level from the level switches (low / medium / high)
`W` icon	Outlined / white box	Tank-wash valve closed / open
Message	The font chosen in Settings → Screen font (from 3.6): its main size when a page fits the width, otherwise the smaller size of the same font; pages change every 3 s (from 3.3)	What the system is doing now, what it means, what to do

Start-up animation (from firmware 3.2)

After power-on or a restart, a Border Collie chases a butterfly across the whole screen for about 4 seconds: it gallops in, crouches and stalks the butterfly, then sprints after it. Then `REUSE WATER` / System ready shows for 2 seconds, then `READY` / *Start washer › Press START*.

- The rest of the controller starts at the same time: Wi-Fi connects and the safety task runs; the animation does not delay anything.
- Notices that arrive during the animation (the IP address, the Wi-Fi setup key, `INTERRUPTED`) are kept and shown after System ready.
- START, or Wash tank in the app, ends the animation at once. Without a working screen it is skipped.
- The drawings are in `BootAnimation.h` (about 1.6 KB).

Screen font (from firmware 3.6)

In the app, **Settings** → **Screen** → **Screen font** chooses the letters of the screen messages. Save the settings and look at the controller screen; try each style from where the screen is usually read, and keep the clearest. The four styles were picked by the household in 3.6 and replace all earlier ones (the six of 3.4, Helvetica Bold included).

Choice (Spanish)	Style	Made from	Main size / smaller size	Messages in the main size
Hyperlegible — *Hiperlegible* (default)	Open bold sans, made for readers with low vision	Atkinson Hyperlegible Bold	13 / 12 px	61 of 91
Plain sans — *Sencilla*	Plain bold sans, the largest that fits most messages	PT Sans Bold	14 / 13 px	84 of 91
Terminal bold — *Terminal gruesa*	Pixel font drawn for screens, tall letters	Terminus Font Bold	16 / 12 px	45 of 91

Choice (Spanish)	Style	Made from	Main size / smaller size	Messages in the main size
Schoolbook bold — *Escolar gruesa*	Serif pixel font, like a school reader	Adobe New Century Schoolbook Bold	14 / 11 px	49 of 91

A part that does not fit the 128 px line in the main size is shown in the smaller size of the same style, which every message fits, so the letters never change style between pages. The top line keeps its small built-in font. The change shows on the screen as soon as the settings are saved, and it is kept after a restart. After the upgrade from 3.4 or 3.5 the screen starts in Hyperlegible, because the earlier styles were removed; a history file from before 3.6 restores every setting except the screen font. Font origins and licences: `licenses/README.md` in the firmware folder (free fonts under the SIL Open Font License, and the Adobe pixel font).

Messages

English, then Spanish. Pages are separated by ›; a page in brackets appears only when it applies. From 3.4 the top line says the step or the event and the message only adds what is new (for example `DONE` shows only the litres).

Top line	Message pages	What it means	Action needed
Full screen: a dog chasing a butterfly		The controller has just switched on (about 4 s)	None; START skips it
`REUSE WATER` `REUSO DE AGUA`	System ready Sistema listo	Start-up finished (2 s)	None
`READY` `LISTO`	Start washer › Press START Inicie lavadora › y pulse START	Waiting for a new cycle; the tank icon shows the tank level	Start the machine’s program, then press START
`STARTING` `INICIANDO`	Getting ready... Preparando...	START was pressed	None
`WASH 1 OF 4` `LAVADO 1 DE 4`	Waiting to fill (› 25 min left) Esperando agua (› Faltan 25 min)	The machine has not started taking water yet	Start the machine’s program if it is not running
`WASH 1 OF 4`	Reusing water Reusando agua	The machine is filling with water saved in the tank; the tank is being cleaned (`W` lights every 10 s)	None
`WASH 1 OF 4`	Using tap water Agua de la llave	The machine is filling from the tap	None

Top line	Message pages	What it means	Action needed
`WASH 1 OF 4`	Tank low › Using tap water Tanque bajo › Agua de la llave	The tank has too little water, so tap water is used	None
`WASH 1 OF 4`	Washing... (› 25 min left) Lavando... (› Faltan 25 min)	The machine has filled and is washing	None
`DRAIN 2 OF 4` `DESAGUE 2 DE 4`	Draining › soapy water Desaguando › agua jabonosa	Soapy wash water goes to the drain, not the tank	None
`RINSE 3 OF 4` `ENJUAGUE 3 DE 4`	Using tap water Agua de la llave	Rinse fill from the tap	None
`RINSE 3 OF 4`	Rinsing... Enjuagando...	The machine has filled and is rinsing	None
Steps 1, 2 or 3	No water flow › Check washer › Call installer Sin flujo › Revise lavadora › Pida ayuda	The step has waited longer than the step time limit for water flow	Check the washing machine; if it runs, press STOP and check the Service tab of the app
`SAVE 4 OF 4` `AHORRO 4 DE 4`	Filling tank › Ends in 18:42 Agua al tanque › Fin en 18:42	Rinse water is stored; the clock shows when the cycle ends	None
`SAVE 4 OF 4`	Low flow › To the drain › Ends in 18:42 Flujo bajo › Al desague › Fin en 18:42	The drain flow is at or below the minimum flow to save, so the fill valve is closed and the rest goes to the drain (from 3.4)	None; saving starts again if the machine pumps out again
`SAVE 4 OF 4`	Tank full › Ends in 18:42 Tanque lleno › Fin en 18:42	The tank is full; the rest goes to the drain	Use or empty the tank if this happens often
`TANK CLEANING` `LIMPIEZA TANQUE`	Until full › 9:42 left Hasta llenarse › Quedan 9:42	Manual tank wash; the clock shows the time left before the safety limit	None

Top line	Message pages	What it means	Action needed
`TANK CLEANED` `TANQUE LIMPIO`	Tank full Tanque lleno	The manual wash filled the tank	None
`WASH STOPPED` `LAVADO DETENIDO`	Tank not full Tanque no lleno	Stopped from the app, the STOP button, or by a cycle starting	None
`WASH STOPPED`	Check tank › Call installer Revise tanque › Pida ayuda	The time limit was reached, or the level switches disagree	Check both float switches and the wash water supply
`INTERRUPTED` `INTERRUMPIDO`	Check washer › Start again Revise lavadora › Inicie de nuevo	The controller restarted during a cycle (15 s)	Check the washing machine; start again if needed
`ReuseWater–Setup`	Wi-Fi setup › Key gw-a1b2c3 Configurar Wi-Fi › gw-a1b2c3	The Wi-Fi setup page is open; the top line is the network to join and the key is its password	Join it on a phone with that key
IP address	Wi-Fi connected › App: this IP Wi-Fi conectado › App: esta IP	Connected to the home Wi-Fi; the top line is the app's address (6 s)	Note it, or use `reusewater.local`
`UPDATING` `ACTUALIZANDO`	Do not unplug No desconecte	A wireless firmware update is running	Wait; do not switch off
`UPDATED` `ACTUALIZADO`	Restarting Reiniciando	The update finished; the controller restarts	None
`UPDATE FAILED` `NO ACTUALIZADO`	Try again Reintente	The wireless update did not finish; the old firmware still runs	Try the upload again
`STOPPED` `DETENIDO`	Cycle cancelled Ciclo cancelado	STOP was pressed	Press START to run again
`DONE` `TERMINADO`	Reused 26 L › Saved 30 L (or Cycle complete) Reusados 26 L › Guardados 30 L (o Ciclo completo)	The cycle finished: tank water used instead of tap water, and rinse water stored for the next wash	None

Brightness (from 3.4): the screen runs at full brightness (contrast 255; the display library starts at 207), so the letters are sharper from a distance.

Minutes left (from 3.3): once every step has been timed at least once, steps 1–3 add a page *25 min left* / *Faltan 25 min*. It is the machine's typical time for the rest of the current step (at least 1 minute), the typical times of the later steps, and the fixed length of step 4. Until then the page is left out.

A notice (for example `DONE`) stays on long enough to show each of its pages once (3 s per page), and never less than its own time (4 s; 15 s for `INTERRUPTED`).

Every page fits the screen without scrolling: each one is measured in the 12 px font by `tools/check_screen_text.py` (127 texts and pages in both languages), and short pages use the larger 14 px size. A countdown in a page does not restart the pages. The dashboard shows the same status as full sentences.

Short pages are centred and still. A page wider than 128 px would pause for 1.5 s, then scroll at 50 px/s and repeat. The display only sends a new frame when something changes, when the page changes, while text scrolls, or during the start-up animation.

7. Configuration parameters

All timing and behavior settings are `#define`s in `GreyWaterSystem.h`; change them there and re-flash.

Constant	Default	Unit	Controls	Guidance
`OPERATION_DELAY`	250	ms	How often each state updates relays and sensors	Keep 100–1000; other timings are rounded to this step
`TRANSITION_DELAY`	1200000	ms	Length of step 4 (20 min) before returning to Idle	Set to how long the machine's rinse drain lasts
`EDGE_DETECTION_DELAY`	4000	ms	Flow must last this long to count as "flow detected"	Raise if short spurts end steps early
`TANK_WASH_PERIOD`	10000	ms	Interval between tank-wash openings	Multiple of `OPERATION_DELAY`
`TANK_WASH_OPEN_TIME`	2000	ms	Wash valve open time per interval	Less than `TANK_WASH_PERIOD`, else always open
`DEBOUNCE_DELAY`	20	ms	Button debounce	20–50
`CONTRL_SSR_DELAY`	0	ms	Blocking delay after each SSR write	Keep 0; any value blocks `loop()` 5× per update
`DISPLAY_FRAME_MS`	40	ms	Frame interval while scrolling and during the start-up animation (25 fps)	Not below about 20

Constant	Default	Unit	Controls	Guidance
`SCROLL_SPEED_PX`	2	px/frame	Scroll speed (50 px/s)	1–3
`SCROLL_PAUSE_MS`	1500	ms	Rest before a long message scrolls	—
`SCROLL_GAP_PX`	40	px	Gap before a scrolled message repeats	—
`NOTICE_DURATION`	4000	ms	How long "Cycle stopped" / "Cycle complete" stays	—
`BUZZER_PIN`	`D1`	—	Pin of the optional buzzer (D12 until 3.1)	Keep D1 unless the wiring changes; do not use `Serial0`
`DEBUG_MODE`	false	—	Serial debug output at 9600 baud	Set true for troubleshooting
`SENSOR_TAP_TYPE`	YFB1	—	Tap flow sensor model	Must match the fitted sensor
`SENSOR_DRAIN_TYPE`	YFB10	—	Drain flow sensor model	Must match the fitted sensor

The start-up animation speed and stalk length are `INTRO_SPEED_PX` and `INTRO_STALK_FRAMES` in `GreyWaterDisplay.cpp`.

The SSR logic level is set in the `.ino` (`ssrON = HIGH`, `ssrOFF = LOW`). For active-LOW relay boards, swap the two values.

From firmware 2.0, the wash timing, final drain length, flow detection time and alert limits are changed on the dashboard (section 11) and saved in the controller. The `#define` values are only the factory defaults used on first boot.

8. Exceptions and troubleshooting

The firmware handles a low or full tank automatically, but it has no timeouts on steps 1–3, so a missing flow signal leaves the cycle waiting until Stop is pressed; from firmware 2.9 it alerts when a step takes longer than the alert time.

Conditions handled by the firmware

Condition	Detected by	Automatic response	OLED
Tank low during Fill 1/4	Level switches	Fills from the tap instead of the tank	Tank low: tap
No tap demand yet in Fill 1/4	Tap flow sensor	Stays on tap water until flow is confirmed	Using tap water

Condition	Detected by	Automatic response	OLED
Drain flow at or below the minimum flow to save during Drain 4/4 (from 3.4)	Drain flow sensor (2 s average)	Fill valve closes and drain opens on the same 250 ms pass, about 2 s after the flow drops (the measuring time); saving resumes after 2 s above the limit. Logged as *Saving paused* / *Saving resumed*; no phone alert	Low flow
Tank full during Drain 4/4	High-level switch	Stops saving; water goes to the normal drain path	Tank full: drain
High float wet, low float dry for 3 s (from 3.0)	Both level switches	Tank treated as full: no saving, no tank wash, no pump; tap water used. High-priority alert, alarm beeps, event logged	App: Check float switches
Stop pressed	Stop button	Returns to Idle, clears all flags and timers	Cycle stopped
Power loss or reset	—	Restarts in Idle with all outputs off; the cycle is not resumed	Start-up animation, System ready, then Press START (`INTERRUPTED` if a cycle was running)
`millis()` rollover after 49.7 days	—	Timers use unsigned subtraction, so they keep working	—

Faults not detected by the firmware

Problem	Most likely cause	What to check	How to fix	Who
Screen blank, nothing responds	No power, or the display is not detected (from 3.0 the controller keeps running without it and sends an alert)	Power to the board; OLED wires on 3V3, GND, A4, A5	Reseat the OLED wiring; confirm address 0x3C	Technician
Stuck on `WASH 1 OF 4`	One flow sensor reports no water	Washing machine program is running; both flow sensors (D2, D3)	Press STOP; clean or reconnect the sensor	User, then technician
Stuck on `DRAIN 2 OF 4`	Tap flow sensor reports no water	Tap valve opens; tap sensor on D2	Press STOP; check the tap sensor	Technician

Problem	Most likely cause	What to check	How to fix	Who
Stuck on `RINSE 3 OF 4`	Drain flow sensor reports no water	Drain hose and drain sensor on D3	Press STOP; check the drain sensor on the Service tab (pulses counting with no water = noise: fix the wiring); raise the Flow threshold if a steady trickle continues	Technician
Steps change too early	Electrical noise or short spurts of water	Sensor cable routing near mains wires	Separate the sensor cables from mains, pump and valve wiring; fit the diode and pull-up interface; raise the Flow threshold in Settings	Technician
Tank icon wrong	Level switch stuck, or a missing pull-down resistor	Move each float by hand and watch the icon	Clean the floats; fit the 10 kΩ pull-downs	Technician
Fill keeps switching tap ↔ tank	Float bouncing or water sloshing	Float mounting	Mount the float in a calm spot or a still tube	Technician
`W` lights but no tank wash	SSR wired to the wrong pin, or no valve supply	SSR on D0; SSR LED; valve supply	Rewire to D0; restore supply	Technician
Tank-wash valve open the whole fill	`TANK_WASH_OPEN_TIME` is not less than `TANK_WASH_PERIOD`	Settings in the app	Set the open time below the interval	Technician
A valve or the pump never switches	Failed SSR, or SSR needs a different logic level	Voltage on the pin (3.3 V when on); SSR LED	Replace the SSR; swap `ssrON` / `ssrOFF` for active-LOW boards	Technician
Buzzer silent	Buzzer off in Settings, still wired to D12, or a passive buzzer	Settings → Buzzer; signal wire on D1 (from 3.2); module type	Switch it on; move the wire to D1; use an active buzzer module	Technician
Buzzer chirps at power-on	Normal: the chip's start-up messages go out on D1 (TX0)	—	None; it stops as soon as the firmware starts	—

Problem	Most likely cause	What to check	How to fix	Who
Screen stutters, buttons slow	I2C too slow, or a delay added to SSR switching	`Wire.setClock(400000)`; `CONTRL_SSR_DELAY`	Keep 400 kHz and a 0 ms delay	Technician
A new cycle starts by itself	START button held or stuck	Button and its wiring	Release or replace the button	User

Diagnostic procedure

1. Set `DEBUG_MODE` to `true` in `GreyWaterSystem.h` and flash.
2. Open the Serial Monitor at 9600 baud.
3. Watch the state entries (`Entering state: …`), flow rates, tank level and SSR switching while you reproduce the fault.
4. Set `DEBUG_MODE` back to `false` for normal use; the debug output slows the loop.

Developer note: `DEBUG_PRINT(x)` expands to an `if` without braces. Do not use it as the only statement before an `else`.

9. Commissioning and test checklist

Run these checks after installation, after any wiring change, and after flashing new firmware.

Before power-up

- [ ] Every SSR input is wired to its pin from section 2, with a common GND
- [ ] SSR load side and valve/motor supply are fused and switched off
- [ ] Level switches have pull resistors; buttons go to GND
- [ ] Flow sensor signal levels are safe for the board's inputs
- [ ] If a buzzer is fitted: active buzzer module, signal on D1 (not D12)

Power-up and idle

- [ ] OLED shows the Border Collie chasing the butterfly (about 4 s), then `REUSE WATER` / System ready for 2 s, then `READY` / *Start washer › Press START*, with no `DONE`
- [ ] Pressing START during the animation ends it at once
- [ ] Messages read clearly in the bold font at the normal viewing distance; each page stays about 3 s, long enough to read

- [ ] All SSRs are off for the first moment, then tap and drain SSRs turn on in Idle
- [ ] Tank icon follows the level switches when each float is lifted by hand (empty, half, full)
- [ ] If a buzzer is fitted: at most a short chirp at power-on, then silent

Buttons

- [ ] Start in Idle changes the header to `STARTING`, then `WASH 1 OF 4`
- [ ] Stop during each step shows `STOPPED` / *Cycle cancelled* for 4 s, then `READY`
- [ ] Start during a running step has no effect

Full cycle

- [ ] Fill 1/4 with the tank low uses tap water; the `W` icon stays outlined
- [ ] Fill 1/4 with water in the tank switches to tank water once the machine draws water; the motor runs and `W` turns white for 2 s every 10 s
- [ ] Drain 2/4 shows *Draining › soapy water*; the fill valve stays off
- [ ] Rinse 3/4 uses tap water (*Using tap water*); the wash valve stays closed
- [ ] Drain 4/4 saves water (fill valve on, drain off) and the countdown decreases each second
- [ ] With the tank full, Drain 4/4 shows *Tank full › Ends in …*
- [ ] After 20 minutes the OLED shows `DONE` / Cycle complete, then `READY`; with the buzzer on, two short beeps

Faults

- [ ] Disconnect the OLED and reset: the controller keeps running and the app's Service tab shows the screen as not working
- [ ] Disconnect a flow sensor in Fill 1/4: the step waits, and Stop still returns to Idle

10. Maintenance and firmware deployment

Maintenance

Task	Interval	Notes
Check the OLED tank icon against the real tank level	Monthly	Catches stuck or fouled level switches
Clean flow sensors and their inlet screens	Every 3 months	Lint in grey water blocks the YF-B10 drain sensor first

Task	Interval	Notes
Inspect the tank and its wash nozzle	Every 3 months	Adjust the wash valve open time in Settings if sediment builds up
Check valves and SSRs switch (run the commissioning "Full cycle" checks)	Every 6 months	Listen for each valve click; check SSR LEDs
Check connectors and cable strain relief	Every 6 months	Damp environment

Intervals are suggested starting points; adjust them to the water quality and how often the machine is used.

Firmware build and installation

1. Install the Arduino IDE (or arduino-cli) and the board package **esp32 by Espressif Systems**; select **Arduino Nano ESP32**.
2. Install the libraries: Adafruit GFX, Adafruit SSD1306, AgileStateMachine and WiFiManager (from firmware 2.4 the FlowSensor library is no longer needed; the 3.2 screen fonts are part of the sketch).
3. Keep the whole `FSM_WashingMachineSystem` folder together: the `.ino` and every `.cpp` and `.h` file in it. The folder name must stay `FSM_WashingMachineSystem`.
4. Select the board and port, then Upload (over USB, or wirelessly, see section 11).
5. Run the checks in section 9.

Software modification reference

- Pins, timings and display settings: `GreyWaterSystem.h` (sections 2 and 7).
- Relay combinations and interlocks: `GreyWaterOutputs.h` / `.cpp` (section 3).
- OLED text: `displaySetStatus()` calls in each state's logic function, and the strings returned by `checkAndFillWashingMachine()` and `checkAndFillWaterTank()`. Every screen text is a `TRS("English", "Español")` pair; run `tools/check_screen_text.py` after changing one.
- Screen layout, fonts and start-up animation: `GreyWaterDisplay.cpp`, the `Font*.h` files (made by `tools/make_screen_fonts.py`), `BootAnimation.h`.
- States and transitions: `setupStateMachine()` in the `.ino` (section 5).

11. Wi-Fi features (firmware 2.0)

Firmware 2.0 adds a phone dashboard, alerts, water statistics and wireless updates; the wash cycle runs exactly the same when Wi-Fi is missing. From firmware 3.2 the app is called **Reuse**

Water System (Spanish: *Sistema de Reúso de Agua*).

First-time Wi-Fi setup

1. On first boot, after the start-up animation, the screen shows `ReuseWater-Setup` and the setup key, for example `Key gw-a1b2c3` (in Spanish, only the key). The key is unique to each board (from firmware 3.0) and is also printed on the USB serial monitor.
2. On a phone, join the Wi-Fi network **ReuseWater-Setup** with that key. Only if the screen is missing or broken is the key the old shared password `greywater`.
3. A setup page opens (otherwise browse to `192.168.4.1`). Choose the home network and enter its password.
4. The screen shows the controller's IP address for 6 seconds.
5. Open `http://reusewater.local` (or that IP address) and sign in as `admin`, password `greywater`.
6. In **Settings**, set new passwords and an alert topic (see Phone alerts).

The setup page closes after 3 minutes and the system then runs offline. To open it again, for example after changing the router, hold **STOP for 5 seconds** while the screen shows `READY`.

From firmware 3.0 the setup page only sets the Wi-Fi network and shows information: it cannot install firmware or erase the controller, because it has no password of its own.

> **Name change (3.2):** until firmware 3.1 the app was at `greywater.local`, the setup network was `GreyWater-Setup` and the update port was `greywater`. After the 3.2 update they are `reusewater.local`, `ReuseWater-Setup` and `reusewater`. The IP address, passwords, settings and water history do not change; update any bookmark or home-screen shortcut.

Dashboard

Tab	For	Shows
Overview	Household	What is happening now, using the same words as the controller’s screen; Start cycle and Stop cycle; the storage tank (level, water age, wash valve) and the Wash tank button
Water use	Household	Last wash (when, how long, litres reused and saved); this month’s share of washing water that came from the tank, compared with last month; washes, reused, saved and tap water this month; totals by period
Service	Installers and technicians	Flow sensors (rate, flowing or not, pulse counts); valves and pump; tank level switches; current step with litres taken and the “machine is filled” guard; Wi-Fi, IP address, uptime, alerts and firmware
Settings	Everyone	All the values in the settings table, a test-alert button and a restart button

View-only user (from firmware 3.5): for people who should follow the washes and the water figures but must not change the controller. As `admin`, open **Settings** → **Security**, switch on **View-only user** (*Usuario de solo consulta*), enter a **New viewer password** (8–32 characters, not the dashboard password) and press Save settings. Give that person the user name `viewer` and that password.

- Signed in as `viewer`, the app shows only **Overview** and **Water use**. There is no Start cycle, Stop cycle, Wash tank, *OK, I checked the machine*, Restore to controller or Restore from file, and no Service or Settings tab. A note under the status says changes need the admin sign-in. Download history file still works.
- The controller itself refuses every change, the settings and the event log from `viewer`, so the limit holds even outside the app.
- To take the access away, switch View-only user off and Save: the viewer password stops working at once. To change it, enter a new one and Save.
- A browser remembers one sign-in per address. To try the viewer on your own phone, use a private browser window.
- The STOP button on the controller always works, whoever is signed in to the app.

On the Service tab, the tank-wash valve state is reported by the controller. The other valve and pump states are the expected states for the current step and fill source.

Refresh (from firmware 2.8.1): the app updates itself every 2 seconds. The badge in the top bar shows the connection:

- *Online* with the controller's time: updates are arriving.
- *Updating slowly*: no update for 6 seconds, usually weak Wi-Fi.
- *Offline*: the controller cannot be reached.

Press **Refresh** next to the badge to fetch fresh data at once. It cancels any update that is still waiting.

Keeping the water history

The Water use history survives firmware updates. From firmware 2.8 it is protected in three places:

Where	What it does
The controller	Stores the history twice (main record and backup copy), saves it again before every wireless update, and never overwrites a record it cannot read
Every phone that opens the app	Keeps its own copy, updated automatically. If the controller ever comes back without history, the Water use tab shows the phone's copy and a Restore to controller button

Where	What it does
A history file	Download history file on the Water use tab. Restore from file brings it back, even onto a replaced or fully erased controller

If the Water use tab is empty after an update:

1. Open the app on a phone that was used before the update. Use the same address as usual: `reusewater.local` and the IP address keep separate copies (before 3.2 the address was `greywater.local`).
2. On the Water use tab, press **Restore to controller**. If no phone has a copy, use **Restore from file** with a downloaded history file.
3. Check the Service tab: Water history should read Restored from the app.

Avoid erasing the controller's memory: a normal upload keeps the history. A full flash erase removes it, together with the settings and the Wi-Fi network. That covers Burn Bootloader, recovery mode, uploading with another board or partition scheme selected, and Erase All Flash. Download a history file before any such work.

Safety and service features (firmware 2.9)

Feature	What happens	Where to see it
Interrupted cycle	If the power goes off, the program crashes or the watchdog restarts the controller during a cycle, the controller comes back in Ready with all outputs safe. It shows `INTERRUPTED` / Check machine (after the start-up animation), sends a high-priority alert and logs the event. The cycle is not resumed.	Screen, phone alert, Overview notice (until OK or the next Start), Service → Recent events
Watchdog	If the program freezes for 5 s, the controller restarts with every output off	Service → Last restart: Watchdog (program froze)
Wireless updates	Accepted only while the controller is Ready and no tank wash runs. During a cycle, the Arduino IDE's network upload gets no answer	Service → Wireless updates
Valves and pump	The Service tab shows what each output is actually set to	Service → Valves and pump
Event log	The last 50 events since the controller started: steps, machine filled, pump on and off with the reason, alerts, tank wash, Wi-Fi, updates, restores	Service → Recent events
Password protection	The app asks for new passwords while the defaults are in use. After 5 wrong passwords in a minute from one device, it refuses that device for a minute; other devices keep working (per device from 3.0)	App

Feature	What happens	Where to see it
Buzzer (optional)	An active buzzer module on D1 / TX0 (D12 until 3.1), switched on in Settings: two short beeps when a cycle ends, three long beeps on alarms	Settings → Cycle
Typical step times	The controller learns how long steps 1–3 usually take; the app shows “Usually about 25 min”	Overview

A printable **Quick Card** for the household, with the installer’s commissioning checklist, is a separate document.

Firmware 3.0 to 3.2

Feature	What happens	Where to see it
Safety task (3.0)	Every 20 ms, independent of the app and Wi-Fi: reads the floats, stops the pump within about 0.3 s when the machine stops drawing, keeps the fill and tank-wash valves closed while the high float is wet	Service → Valves and pump; Recent events
Float-switch fault (3.0)	High float wet with the low float dry for 3 s: the tank is treated as full, the pump is not used, an alert is sent and the alarm beeps	Alert *check level switches*; app *Check float switches*; Service → Tank level switches
No screen (3.0)	If the screen does not answer at start-up, the controller runs without it and sends an alert	Service → Screen
Confirmed actions (3.0)	The app shows Cycle started, Cycle stopped or Tank wash started only after the controller confirms; Send a test alert reports whether the alert was delivered	App messages; Service → Last alert
Loop time and memory (3.0)	The slowest program pass and the free memory, to spot problems early	Service → Slowest loop pass, Memory
Language (3.1)	Settings → Language: Español (default) or English, for the app, the screen and the phone alerts. The event log stays in English for service	Settings
Screen font (3.4, new styles in 3.6)	Settings → Screen → Screen font: four letter styles for the messages on the controller screen (Hyperlegible, Plain sans, Terminal bold, Schoolbook bold); a long part uses the smaller size of the same style	Settings; the controller screen
New name (3.2)	Reuse Water System (*Sistema de Reúso de Agua*): app title, screen, alert titles (*Reuse Water: …* / *Reúso de Agua: …*), `reusewater.local`, `ReuseWater-Setup`, update port `reusewater`. Settings, history, backups and passwords are kept	App, screen, alerts
Start-up animation (3.2)	A Border Collie chases a butterfly across the screen for about 4 s after power-on, then System ready; START ends it	Screen

Feature	What happens	Where to see it
Clearer screen font (3.2)	Messages in Helvetica Bold, 14 px when they fit and 12 px otherwise; larger tank icon in the header	Screen
Buzzer on D1 (3.2)	The optional buzzer moved from D12 to D1 (TX0); a short chirp at power-on is normal	Wiring; Settings → Buzzer

Spanish screen texts (3.1): the screen fonts have no accents, so the screen shows short plain words, for example `LISTO` / *Inicie lavadora › y pulse START*, `LAVADO 1 DE 4`, `DESAGUE 2 DE 4`, `ENJUAGUE 3 DE 4`, `AHORRO 4 DE 4`, *Reusando agua › del tanque*, *Agua de la llave*, *Ahorrando agua*, `TERMINADO` (texts from 3.3), and at start-up `REUSO DE AGUA` / Sistema listo. The app shows the full Spanish sentences. The Wi-Fi setup page itself stays in English (it comes from the WiFiManager library).

Manual tank wash

Use it to rinse the storage tank between washes. The wash valve opens and closes by itself as soon as the high-level switch reports that the tank is full.

1. Check that the dashboard shows **Ready** and the tank is not full.
2. In the **Storage tank** panel, press **Wash tank**. The screen on the controller shows `TANK CLEANING` with *Until full › 9:42 left*, the time left before the safety limit.
3. Wait. When the high-level switch reports water, the valve closes and the dashboard shows Tank full: wash valve closed.
4. To end it early, press **Stop washing** on the dashboard or **STOP** on the controller. Starting a cycle also ends it.

The wash ends because	The valve	Screen and dashboard	Phone alert
High-level switch reports full	Closes at once	Tank full: valve closed	Tank wash complete
High-level switch wet, low-level switch dry	Closes at once	Level switches disagree: check sensors	Check level switches (high priority)
Safety time limit reached (default 10 min)	Closes	Time limit: check level switch	Tank wash time limit (high priority)
Stop washing, STOP button, or a cycle starts	Closes	Tank wash stopped	None

The button is unavailable while a cycle runs or when the tank is already full.

Phone alerts

Install the free **ntfy** app, subscribe to a long, hard-to-guess topic name, enter the same topic in Settings and press **Send a test alert**. From firmware 3.2 every alert title starts with *Reuse Water:* (Spanish: *Reúso de Agua:*), for example *Reuse Water: tank full*.

Alert	Sent when	Priority
Cycle complete	Step 4 finishes; includes litres reused and saved	Low (silent on most phones)
Cycle stopped	STOP pressed on the controller or the dashboard	Minimum (silent)
Tank full	The tank fills during step 4 and extra water goes to the drain	Low
Check sensors	Step 1, 2 or 3 runs longer than the alert time (default 90 min) without the expected flow	High
Tank water is old	Stored water is older than the limit (default 48 h)	Normal
Tank wash complete	A manual tank wash filled the tank and the valve closed	Low
Check level switches	A manual wash stopped because the high-level switch is wet but the low-level switch is dry	High
Tank wash time limit	A manual wash reached its time limit before the tank reported full	High
Tank pump not pumping	The tank pump started 3 times in one cycle without any water flowing; tap water was used instead	High
Cycle interrupted	The controller restarted during a cycle (power cut, crash or watchdog)	High

Routine news is low priority so people don't mute the topic; problems are high priority so they are noticed.

The topic may use only letters, numbers, `-` and `_` (no spaces or accents); Settings refuses anything else. Leave *Alert server* as `https://ntfy.sh`; if the subscribe link (`https://ntfy.sh/your-topic`) is pasted there, the controller removes the topic. Tapping an alert opens the app (from 3.3).

From firmware 3.3.1, an alert that cannot be sent (alert server down or busy, no internet, no Wi-Fi) is kept and tried again for up to 30 minutes: after 5 s, 10 s, 20 s and so on, then every 5 minutes. Up to 6 alerts can wait at the same time, each on its own schedule. *Recent events* on the Service tab shows *Alert failed (code n), trying again for up to 30 min* and later *Alert delivered (try n)* or *Alert failed*. The test alert gives up after 30 seconds, so the app can answer while you wait; if the alert server is down it reports *could not connect*. (3.3.0: 3 tries and 5 minutes for Wi-Fi; until 3.2.0:

one try and 60 seconds for Wi-Fi.) Anyone who knows the topic name can read the alerts, so keep it private.

Wireless firmware updates

1. Stop any running cycle.
2. In the Arduino IDE, choose **Tools > Port** and the network port named `reusewater` (a board still running 3.1 or earlier shows `greywater`).
3. Upload and enter the update password.
4. The controller switches every valve and the pump off, shows `UPDATING`, then restarts with the start-up animation.

Settings on the dashboard

Setting	Default	Allowed range
Wash every	10 s	1–600 s
Keep the valve open	2 s	0.25 s up to the interval minus 0.25 s
Manual wash time limit	10 min	1–30 min
Final drain length (step 4)	20 min	1–120 min
Flow detection time	4 s	1–30 s
Flow threshold	1 L/min	0.2–10 L/min; water counts as flowing above it and as stopped below it for 2 s
Minimum fill	3 L	0.5–50 L; water the machine must take before a drain can end Fill 1/4 or Rinse 3/4
Minimum flow to save	0.5 L/min	0.5–10 L/min; in Drain 4/4 rinse water goes to the tank only while the drain flow is above this; at or below it the fill valve closes and the drain opens (from 3.4)
Start from the dashboard	On	On / off
Buzzer	Off	On / off; needs an active buzzer module on D1 (TX0)
Warn when a step takes longer than	90 min	10–600 min
Warn when tank water is older than	48 h	1–240 h

Setting	Default	Allowed range
Alert server / topic	https://ntfy.sh / empty (alerts off)	Create a private topic fills in a random topic and shows the subscribe link
Time zone	UTC–5 (`<-05>5`)	Chosen from a list of common zones; Other accepts a POSIX code
Language	Español	Español / English, for the app, the screen and the alerts (from 3.1)
Screen font	Hyperlegible	Four letter styles for the screen messages, see *Screen font* in section 6 (from 3.4; styles of 3.6)
Dashboard and update passwords	`greywater` (the app asks you to change it)	8–32 characters; the default cannot be chosen again; both take effect at once (from 3.0)
View-only user and viewer password	Off	User `viewer`, sees only Overview and Water use; password 8–32 characters, not the dashboard password; off forgets the password (from 3.5)

Wi-Fi troubleshooting

Problem	What to check	How to fix
`reusewater.local` does not open	Phone on the same Wi-Fi; some Android phones do not support .local names; an old bookmark to `greywater.local`	Use the IP address shown on the screen at start-up, or from the router's device list; update old bookmarks
Dashboard says it can't reach the controller	Controller powered; Wi-Fi signal (shown in the page footer)	Press Refresh in the top bar. If it stays Offline or Updating slowly, move the router or add a repeater; below about –80 dBm the link drops
Test alert: *could not connect*	The controller reached the internet but not the alert server; the public ntfy.sh server is sometimes down for hours	Open `https://ntfy.sh` on a phone on the same Wi-Fi. If it does not load either, wait and try later: real alerts are tried again for 30 minutes. If it loads, restart the controller
No alerts arrive	Topic in Settings matches the app exactly; footer says alerts are on; Service → Last alert and Recent events give the reason	Press Send a test alert. *The server refused the topic*: use only letters, numbers, - and _. *Too many alerts*: wait a few minutes. *No Wi-Fi / could not connect*: check the router's internet. Also check the phone's notification permission and battery saver for the ntfy app
Cycle complete or Cycle stopped alerts arrive without sound	They are sent at low and minimum priority on purpose	In the ntfy app, open the topic's notification settings to change how low-priority alerts are shown

Problem	What to check	How to fix
Network changed	—	Hold STOP for 5 s while `READY`, join `ReuseWater-Setup` with the key shown on the screen, and set the new network
Wireless upload fails	Network port listed in the IDE (`reusewater`); update password	Retry; if it keeps failing, upload once over USB
Water age shows No water stored with water in the tank	The tank level must be above the low switch for 30 s	Check the low-level switch

Security

- Change both default passwords on first use.
- Keep the dashboard on the home network; do not forward router ports to it.
- Turn **Start from the dashboard** off in Settings if a cycle could be started while nobody is at the machine.
- The Wi-Fi setup network uses a key unique to each board, shown on the screen while the setup page is open (from firmware 3.0).

Appendix A. Abbreviations

Term	Meaning
EV	Electrovalve (solenoid valve)
FSM	Finite state machine
GND	Ground
I2C	Two-wire serial bus used by the OLED (SDA data, SCL clock)
ISR	Interrupt service routine
OLED	Organic LED display
RCD / GFCI	Residual-current device / ground-fault circuit interrupter
SSR	Solid-state relay
TX0 / RX0	The board's serial transmit / receive pins (UART0), shared with D1 / D0

Appendix B. Revision history

Version	Date	Description
3.6	2026-11-04	Four new screen fonts picked by the household: Hyperlegible (default), Plain sans, Terminal bold, Schoolbook bold; a long part uses the smaller size of the same font instead of Helvetica 12 px. The six earlier styles, Helvetica Bold included, are removed; the screen starts in Hyperlegible after the upgrade
3.5	2026-11-04	View-only user: a second sign-in, `viewer`, with its own password set in Settings → Security. It sees only Overview and Water use and the controller refuses its changes; off by default
3.4.1	2026-11-04	Shorter screen messages that do not repeat the top line (for example *Desaguando › agua jabonosa*, `TERMINADO` *Reusados 26 L › Guardados 30 L*); full screen brightness
3.4	2026-11-04	Minimum flow to save (default 0.5 L/min): in step 4 a drain flow at or below it closes the fill valve and opens the drain at once, and a new pump-out is saved again; screen *Low flow › To the drain*. Screen font chosen in Settings → Screen: original Helvetica Bold or five 13 px styles (narrow bold, narrow light, semi-narrow, round, classic serif), applied at once and kept after a restart; settings version 8 (3.3 settings are kept)
3.3.1	2026-11-03	Phone alerts that cannot be sent (alert server down, no internet or Wi-Fi) are tried again for up to 30 minutes, each on its own schedule; the test alert still answers within 30 s. The app's subscribe link no longer repeats a topic pasted into Alert server
3.3	2026-11-03	Screen messages explain themselves in short pages that change every 3 s (what is happening, what it means, what to do); steps named in words (`WASH 1 OF 4`); minutes left from the second wash; tank level on the Ready screen; litres reused and saved at the end. Phone alerts: correct failure reason in the app (codes no longer clash with HTTPClient's); up to 3 tries per alert and 5 min wait for Wi-Fi; 10 s network timeouts; topic checked (letters, numbers, - and _); pasted subscribe link removed from the server address; tapping an alert opens the app
3.2	2026-10-30	New name Reuse Water System (`reusewater.local`, `ReuseWater-Setup`, update port `reusewater`, alert titles); Border Collie start-up animation; Helvetica Bold screen font (14 / 12 px); larger tank icon; optional buzzer moved from D12 to D1 (TX0); Runbook and Quick Card rebuilt from editable sources
3.1	2026-10-30	Spanish (Colombia) for the app, the screen and the phone alerts, chosen in Settings → Language (Spanish by default); alerts sent as JSON so accents arrive intact; setup key shown on the screen
3.0	2026-10-30	Safety task and output interlocks; float-switch rules, debouncing and fault alert; runs without a screen; per-board Wi-Fi setup key and no firmware upload on the setup page; update password applied at once; lockout per device; app reports only confirmed actions and alert delivery; compressed app page; code split into modules. Tested on the installed system

Version	Date	Description
2.9	2026-10-29	Interrupted-cycle detection and alert; watchdog; wireless updates blocked during a cycle; real valve and pump states and an event log on the Service tab; password prompt and lockout; settings in the history file; time zone list; private alert topic; quieter routine alerts; typical step times; optional buzzer on D12; full-cycle simulation tests; Quick Card
2.8.1	2026-10-29	Refresh button in the app's top bar; requests time out so a slow connection no longer freezes the screen; "Updating slowly" badge after 6 s without an update
2.8	2026-10-29	Water history is never lost: stored twice on the controller, unreadable records kept aside, saved before wireless updates; each phone keeps a copy and can restore it; history file download and restore; history source shown on the Service tab
2.7	2026-10-21	Controller screen rewritten for households, all messages fit without scrolling; app reorganised into Overview, Water use, Service and Settings tabs
2.6	2026-10-21	"Machine is filled" guard (fixes the early move to Drain introduced in 2.4); Minimum fill setting; settings migrate without reset
2.5	2026-10-21	Tank pump protection: the pump stops within 0.3 s when the machine closes its inlet; dry-start alert
2.4	2026-10-21	Drain-sensor fault fixed: new flow meter; diode and pull-up sensor wiring; FlowSensor library removed; accessibility fixes
2.3	2026-10-21	Dashboard shows results: This month panel, last wash summary, tank readiness
2.2	2026-10-21	Manual tank wash that stops when the tank is full
2.1	2026-10-21	Redesigned dashboard with Start and Stop buttons; tank-wash valve moved to D0
2.0	2026-10-21	Wi-Fi: dashboard, phone alerts, statistics, wireless updates, settings
1.2	2026-10-19	Tank-wash valve moved from D12 to D1
1.1	2026-10-19	Quick start, pinout drawing, wiring diagrams, clearer tables
1.0	2026-10-19	Initial release